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Patterns of Shared Genetic Risk Between Chronic Pain, Psychopathologies, and Neuroticism

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Abstract

Mounting evidence supports a prominent neurobiological and psychological component to chronic pain. However, the genetic association between chronic pain, psychiatric conditions, and neuroticism remains unclear. To assess this, we estimated genetic correlations between latent factors for General and Musculoskeletal Pain and Externalizing, Psychotic Thought, Compulsive Thought, and Internalizing psychopathology using genomic structural equation models of published genome-wide association studies. We further estimated the proportion of each pain-psychopathology correlation explained by neuroticism. We observed substantial genetic correlations for the General Pain factor with Internalizing and Externalizing psychiatric factors ($r = .7$ for both); genetic correlations with Psychotic and Compulsive Thought disorders were negligible ($r < .2$). Neuroticism explained substantial (19%–53%) shared genetic variance for General Pain with Externalizing and Internalizing factors. Overlapping genetic risks for chronic pain, psychiatric conditions, and neuroticism suggest shared biological mechanisms, underscoring the importance of chronic-pain assessment and treatment programs that address these shared mechanisms.

Keywords

genomic SEM, chronic pain, externalizing, internalizing, neuroticism, open data

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Chronic pain, traditionally viewed as a biomedical condition, is now understood as a condition with a pronounced psychiatric component. As evidence of this component, there are numerous dimensions of overlap in symptoms and diagnoses of pain and mental disorders. Pain and psychiatric conditions correlate at both phenotypic (Elliott et al., 2003; McGeary et al., 2016; Walid & Zaytseva, 2009) and genetic (Freidin et al., 2019; Johnston et al., 2019; McIntosh et al., 2016) levels and share psychological (Dimova & Lautenbacher, 2010; Linton & Shaw, 2011; Nicholas et al., 2011; Uddin et al., 2021; Wise et al., 2010), neurological (Doan et al., 2015; Walker et al., 2014), and behavioral (Booth et al., 2012; Johannsen et al., 2015) risk factors. Moreover, they respond to some of the same drugs (Dharmshaktu et al., 2012; Duan et al., 2018; Krell et al., 2005; Micó et al., 2006) and to the same behavioral (Hearing et al., 2016; Pongan et al., 2017) and psychological (Ashar et al., 2022; Hoffman et al., 2007; Roditi & Robinson, 2011) interventions. In addition, somatic-symptom (or somatoform) disorders are a category of mental-health conditions marked by physical symptoms, including pain (Löwe et al., 2022), and these disorders now

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provisionally comprise one of six overarching spectra of psychopathology in the Hierarchical Taxonomy of Psychopathology (HiTOP; Kotov et al., 2017). These commonalities suggest shared underlying biological mechanisms that are important to jointly understanding and treating both the chronic-pain and psychiatric-condition experiences.

Despite the widespread recognition of overlap between psychiatric and chronic-pain conditions in the research domain, there is consistent evidence of under-assessment and undertreatment of comorbidities in the clinic (Agüera et al., 2010; Kirmayer et al., 1993; Linton & Bergbom, 2011; Woodling et al., 2022). Depression, for example, is a mental-health condition with medically unexplained pain as a prominent symptom. Nevertheless, chronic pain may obfuscate emotional symptoms of depression, leading to underdiagnosis (Jaracz et al., 2016; Katon, 1984), especially given inadequate psychiatric training in practitioners who see pain patients and vice versa (Goesling et al., 2018).

Studies of genetic correlations between chronic pain and mental-health disorders (Chen et al., 2023; Johnston & Huckins, 2023) have been important in identifying the extent of shared risk and, consequently, likelihood of comorbidity between these conditions, providing additional impetus for joint assessment of pain and psychiatric conditions. However, the studies that have examined genetic correlations between psychopathologies and chronic pain have focused on select individual conditions, such as depression and musculoskeletal or abdominal pain (Meng et al., 2020) or anxiety and back pain (Pinheiro et al., 2018). A more extensive sampling of both chronic-pain conditions and externalizing and thought disorders in addition to the commonly studied internalizing conditions is needed to obtain a more comprehensive view of comorbidities that should be considered together. In addition, marginal pairwise correlations do not make patterns of relationships easily accessible. They may even mask underlying relationships by not accounting for a compensatory covariate, as may occur when multiple pathologies contribute to a given presentation of symptoms but only a subset of them is examined (Zorina-Lichtenwalter et al., 2019).

To address these limitations, we had the following two study aims: first, examine shared genetic susceptibility between a wide variety of common chronic-pain conditions beyond pain symptoms included in the somatoform grouping of the HiTOP framework (e.g., headache and gastrointestinal complaints) and a variety of conditions from different psychopathology dimensions, that is, internalizing, externalizing, and thought disorders, and second, use latent genetic factors underlying pain and psychiatric conditions, which capture shared genetic risk, estimated from different samples without the need for comorbidities to be diagnosed in the same individuals (which, among other advantages, solves the problem of underdiagnosed copresenting conditions).

We recently developed a genomic structural equation model (genomic SEM) for chronic pain, with two factors explaining shared genetic variance: a General Pain factor for 24 chronic pain conditions and an orthogonal Musculoskeletal Pain factor that captured remaining covariances among musculoskeletal-pain conditions after accounting for the General Pain factor. We chose this model based on an exploratory–confirmatory factor-analysis approach and after comparing alternative models, such as grouping conditions by anatomic and putative etiologic similarity (Zorina-Lichtenwalter et al., 2023); for a more detailed explanation of model derivation, see the Supplemental Material available online. The power and breadth of conditions assessed in the 500,000-person UK Biobank (UKBB) together with the ability of the genomic SEM to estimate correlations between conditions not diagnosed in the same people made it possible to obtain these pain factors. Likewise, a model of four correlated genetic factors (Externalizing, Internalizing, Compulsive Thought, and Psychotic Thought disorders; Morrison et al., 2023) has been proposed to account for genetic clustering of psychiatric disorders (Caspi et al., 2014). The correlated factors model in Caspi et al. (2014) includes three factors. However, this model was derived from a population-based sample, the Dunedin Multidisciplinary Health and Development Study from New Zealand, and several more severe mental-health conditions, such as anorexia and bipolar disorder, were not included. Taking this model into genetic space and including a wider range of diagnoses from clinical cohorts yielded a four-factor model, in which thought disorders divided into compulsive and psychotic factors with strong wihtin-factor loadings and a moderate interfactor correlation and differential associations with sleep-health measures (Morrison et al., 2023). This splitting of the thought-disorder factor is also consistent with other genomic SEM models of psychopathology that have used similar sets of disorders (Lee et al., 2019).

These models constitute important steps toward comprehensive clinical assessment and treatment programs that consider common causes rather than address each condition in isolation. To advance these goals, we examined genetic correlations between pain and psychiatric factors, given their well-documented comorbidity (Velly & Mohit, 2018). In addition, we assessed genetic overlap for these factors with neuroticism, which has reported associations with psychiatric conditions (Jylhä & Isometsä, 2006; Subica et al., 2016), chronic pain at genetic (Meng et al., 2020) and phenotypic levels (Banozic et al., 2018; Coen et al., 2011; Goubert et al., 2004), and lower pain tolerance (Lynn & Eysenck, 1961). We sought to quantify the extent of shared genetic susceptibility between these conditions that could help characterize the General Pain factor, which is the least well defined among the considered factors. Given that neuroticism is a personality trait characterized by a negative outlook and increased distress from health problems (Costa & McCrae, 1985), the magnitude of genetic correlations offers insight into the contribution of neuroticism to the chronic-pain-disease burden. Furthermore, a model that combines neuroticism with the pain-psychiatric genetic factors will enable an assessment of genetic relationships among all three domains and will permit us to test whether genetic effects shared between neuroticism and pain or between neuroticism and psychopathology account for any genetic risk shared between pain and psychiatric factors, which may have important implications for developing joint somatic and psychological treatment plans for pain management.

Our overarching goal is to characterize the genetic predisposition to chronic pain that operates outside of peripheral injury and tissue characteristics and their attendant psycho-pathophysiological mechanisms. To gain a better understanding of the possible biological overlap that both cross-condition and specific chronic pain may have with mental health, as measured by different psychopathologies and neuroticism, we ask the following questions:

> *Research Question 1:* Which, if any, psychopathology factors correlate most strongly with the two pain factors (General Pain and Musculoskeletal Pain)?
> *Research Question 2:* What is the relationship between neuroticism and each of the genetic factors (two pain factors and four psychiatric factors)?
> *Research Question 3:* Are the relationships between pain and psychiatric factors explained in part by their overlap with genetic risk for neuroticism?

Transparency and Openness

Preregistration

This study was not preregistered.

Data, materials, code, and online resources

Data and scripts for replication of all analyses included in this study are available at https://osf.io/wrqbe/.

Reporting

This study does not involve newly collected data.

Ethical approval

All analyses are based on published genome-wide association study (GWAS) summary statistics without subject-level identifying information. Ethical approval was obtained for the original published studies.

Methods and Materials

Overview

The analysis pipeline, summarized in Figure 1, was as follows: obtain summary statistics on previously done GWASs for pain and psychiatric conditions and neuroticism scores (full scale and two subscales), estimate pairwise genetic correlations for all conditions and neuroticism scores, fit a genomic SEM that combines previously published pain and psychiatric genomic SEMs, and fit the combined genomic SEM with the neuroticism score.

Phenotypes

For 24 pain conditions, 11 psychiatric conditions, and three neuroticism scales, we used association summary statistics from published GWASs. For sample sizes, condition prevalence and heritability, and source publications, see Table S1 in the Supplemental Material.

Pain conditions

Genetic-association summary statistics for pain conditions were taken from GWASs conducted in the UKBB, as described previously (Zorina-Lichtenwalter et al., 2023). Briefly, UKBB study participants, recruited between the years 2006 and 2010 and ages 40 to 69, were extensively phenotyped using questionnaires and hospital and primary-care records. We used data from individuals identified as White Europeans using genomic principal components, as described in Bycroft et al. (2018), and selected the following 24 pain conditions: arthropathies, back pain, chest pain at baseline, chest pain during physical activity, carpal tunnel, chronic widespread pain, cystitis, enthesopathies of the lower limb, enthesopathies elsewhere, gastritis, gout, headache, hip arthrosis, hip pain, irritable bowel syndrome (IBS), knee arthrosis, knee pain, leg pain, migraine, neck/shoulder pain, oesophagitis, rheumatoid arthritis, pain in joint, and stomach pain. These 24 syndromes were either primary pain conditions or conditions with persistent pain as a prominent symptom, as explained further in Zorina-Lichtenwalter et al. (2023). They came from five UKBB categories: medical conditions (100074), health outcomes (713), self-reported medical conditions (1003), health and medical history (100036), and first occurrences (1712). Selection criteria

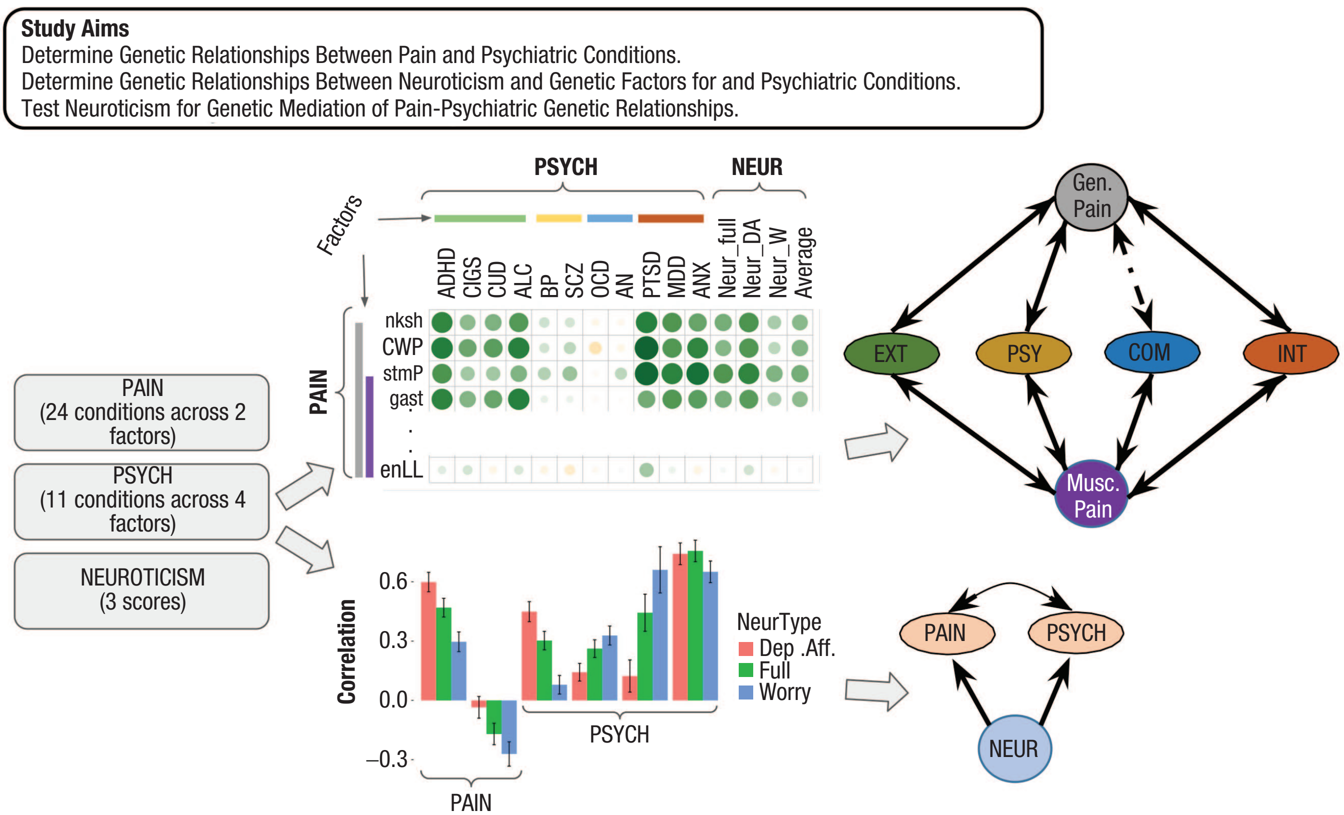

Fig. 1. Overview of aims and methods. GWAS = genome-wide association study; SEM = structural equation model; NEUR = neuroticism; PSYCH = psychiatric condition(s), including four factors: EXT = Externalizing disorders, PSY = Psychiatric disorders, COM = Compulsive disorders, and INT = Internalizing disorders. Pain included two factors: Gen. Pain = General Pain; Musc. Pain = Musculoskeletal Pain.

are further explained in Zorina-Lichtenwalter et al. Sample size varied by condition (see Table S1 in the Supplemental Material) and ranged from 63,982 to 435,971 people.

Psychiatric conditions

The 11 psychiatric conditions came from publicly available results of previously published studies (see Table S1 in the Supplemental Material). All studies consisted of a GWAS meta-analysis of several cohorts. If multiple ancestries were included in the original study, we limited our analyses to European White participants. Conditions included were as follows: anxiety, a study of five types of anxiety disorders: generalized anxiety disorder, panic disorder, social phobia, agoraphobia, and specific phobias (Otowa et al., 2016); major depressive disorder (Howard et al., 2019); posttraumatic stress disorder (PTSD; Nievergelt et al., 2019); alcohol use disorder (Walters et al., 2018); cigarettes per day (CIGS; Liu et al., 2019); cannabis use disorder (E. C. Johnson et al., 2020); attention-deficit/hyperactivity disorder (ADHD; Demontis et al., 2019); schizophrenia (SCZ; Trubetskoy et al., 2022); bipolar disorder (BP; Mullins et al., 2021); obsessive compulsive disorder (OCD; International Obsessive Compulsive Disorder Foundation Genetics Collaborative & OCD Collaborative Genetics Association Studies, 2018); and anorexia nervosa (H. J. Watson et al., 2019). We used the same GWAS summary statistics that were used in Morrison et al. (2023), except for ADHD, BP, PTSD, and SCZ, for which we used more recent studies.

Neuroticism

For neuroticism, we used genetic-association summary statistics (Nagel, Jansen, et al., 2018) from a meta-analysis of data from the UKBB (sample size = 372,903) and Genetic Personality Consortium (sample size = 17,375; De Moor et al., 2012), which analyzed a total of 390,278 participants. In the UKBB, the neuroticism score was a count of "yes" entries in response to the following 12 questions: (1) "Does your mood often go up and down?" (2) "Do you ever feel 'just miserable'?" (3) "Are you an irritable person?" (4) "Are your feelings easily hurt?" (5) "Do you often feel 'fed-up'?" (6) "Would you call yourself a nervous person?" (7) "Are you a worrier?" (8) "Would you call yourself tense or 'highly strung'?" (9)

"Do you worry too long after an embarrassing experience?" (10) "Do you suffer from 'nerves'?" (11) "Do you often feel lonely?" (12) "Are you often troubled by feelings of guilt?" In addition, we used summary statistics for GWASs of two subscales of the UKBB neuroticism scale: depressive affect (Questions 1, 2, 5, and 11) and worry (Questions 6–8 and 10; Nagel, Watanabe, et al., 2018).

Genomic SEM

Using the *genomic SEM* R package, we estimated pairwise genetic covariances for all pain, psychiatric, and neuroticism measures. The genetic correlation r_g (i.e., the standardized covariance; for details, see the Supplemental Material) indicates the magnitude and direction of shared genetic effects on two traits.

To estimate the relationships between genetic factors, we used genomic SEM (Grotzinger et al., 2019). This framework enables us to extract latent factors for pain and psychiatric conditions based on their genetic covariances (in the same way that standard structural equation models use phenotypic covariances). The relationships between the indicators and their latent factors are specified by the researcher with a measurement model (see Figs. S1A and S1B in the Supplemental Material) based on a theoretically or empirically driven hypothesis. We used the published pain (Zorina-Lichtenwalter et al., 2023) and psychiatric (Morrison et al., 2023) models to specify a combined model with two pain factors (General Pain with 24 indicators; Musculoskeletal Pain with 11 indicators) and four psychopathology factors (Externalizing, Internalizing, Psychotic Thought, and Compulsive Thought disorders). Given the prevalence of the two-factor model in adolescent populations and three-factor model in the broader population in psychopathology-epidemiological space (Caspi et al., 2014), readers may wonder about the overall model fit with these variations. Therefore, we provide estimates derived from these models as well. Given that genomic SEM takes as input a matrix of genetic correlations, we report all pairwise genetic correlations for the complete set of 35 conditions plus three neuroticism scores for subsequent analyses. The genetic-correlation matrices for the original pain and psychopathology models are provided, respectively, in Figures S1A and S1B in the Supplemental Material.

Given high correlations among psychiatric factors, we additionally estimated semipartial correlations between the General Pain factor and different combinations of psychiatric factors, controlling for other psychiatric factors. We used Cholesky decomposition (Keenan et al., 2006), a method that decomposes a symmetric matrix, such as the covariance matrix of our genetic factors, and provides estimates of shared and independent variance components. Although comorbidities between psychiatric disorders have previously been modeled using a single-factor model, the p factor (Caspi et al., 2014), there is no strong evidence for the p factor at the genomic level (Grotzinger et al., 2019). Therefore, we did not test our model with the p factor.

To assess the extent to which neuroticism explains associations between the pain and psychiatric factors, we allowed neuroticism to predict both the pain and psychiatric factors and calculated the proportion of the total pain-psychiatric genetic factor correlation attributable to neuroticism. Specifically, for each psychiatric factor, we divided the product of the path coefficients between neuroticism and each factor by the total genetic covariance between General Pain and the psychiatric factor.

Model fit

Because the genetic correlations are based on GWASs with very large sample sizes, the chi-square tests of fit for genomic SEM models, which scale with sample size, are generally large ("overpowered") despite other indications of good fit. We evaluated the model fits using the comparative fit index (CFI), which compares the model fit with one in which the variables are entirely independent, and the standardized root mean square residual (SRMR), a measure of variance unexplained by the model (Hu & Bentler, 1999). A well-fitting model should generally have a CFI $\geq$ .95 and an SRMR $\leq$ .08 (Hu & Bentler, 1999), although lower CFI and higher SRMR thresholds may be acceptable in the context of genomic SEM (Grotzinger et al., 2019).

Results

We estimated genetic correlations of the pain conditions with the psychiatric conditions and neuroticism scores, presented in Figure 2a, which shows the pain conditions in descending order of their loading onto the General Pain factor (Fig. S1A in the Supplemental Material) and psychiatric conditions ordered left to right by loadings onto their respective factors (Fig. S1A in the Supplemental Material). The pain conditions from neck/shoulder pain to knee arthrosis (top 75% of loadings onto the general factor) have the largest genetic correlations with psychiatric conditions and neuroticism scales. Pain-psychiatric condition correlations show patterns segregating by psychiatric factor, and pain-neuroticism correlations are strongest for the full Neuroticism scale and the Depressive Affect subscale, as discussed below. These patterns are consistent with

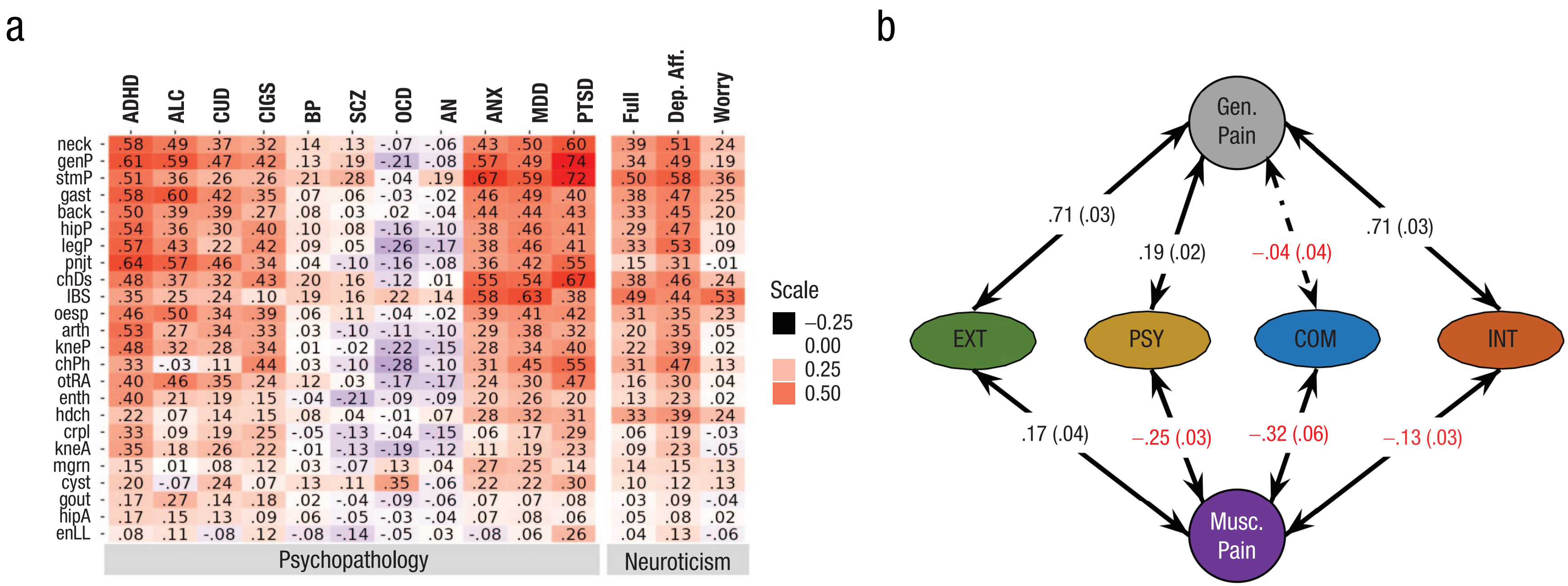

Fig. 2. (a) Genetic correlations estimated in linkage disequilibrium score regression for pain conditions with psychiatric conditions and with neuroticism (full Neuroticism scale, Depressive Affect [Dep. Aff.] and Worry scales). The pain conditions are listed in descending order of their loading onto the General Pain factor, and psychiatric disorders are grouped by their factors and listed left to right in descending order of their loading onto their respective psychiatric factor. (b) Combined genomic structural equation model (SEM) of pain and psychiatric conditions. Interfactor correlations are shown on the arrows. The positive correlations are shown in black, and negative correlations are shown in red. For correlations between psychiatric factors, see Figure S1B in the Supplemental Material available online. In addition to the residual correlations specified for the pain SEM (Fig. S1A in the Supplemental Material), the following residual correlations were estimated (also not shown in figure): CIGS with chest pain at baseline, r_g = .16 (.03), p = 2.7 × 10^(−9), and during physical activity, r_g = .18 (.07), p = 8.6 × 10^(−3); stomach pain with SCZ, r_g = .16 (.03), p = 2.1 × 10^(−8), and with AN, r_g = .22 (.05), p = 2.2 × 10^(−6); IBS with the Compulsive factor, r_g = .33 (.04), p = 4.9 × 10^(−14). Factors: Gen. Pain = General Pain; Musc. Pain = Musculoskeletal Pain; EXT = Externalizing disorders; PSY = Psychotic thought disorders; COM = Compulsive thought disorders; INT = Internalizing disorders. Pain conditions: nksh = neck/shoulder pain; CWP = chronic widespread pain; stmP = stomach pain; gast = gastritis; back = back pain; hipP = hip pain; legP = leg pain; pnjt = pain in joint; IBS = irritable bowel syndrome; kneP = knee pain; arth = arthritis; oesp = oesophagitis; chPh = chest pain during physical pain; rhAt = rheumatoid arthritis; chDs = chest pain; enth = enthesopathies; hdch = headache; crpl = carpal tunnel syndrome; kneA = knee arthrosis; mgrn = migraine; cyst = cystitis; gout = gout; hipA = hip arthrosis; enLL = enthesopathies of lower limb. Psychiatric conditions: ADHD = attention-deficit/hyperactivity disorder; CIGS = cigarettes per day; CUD = cannabis use disorder; ALC = alcohol use disorder; BP = bipolar disorder; SCZ = schizophrenia; OCD = obsessive compulsive disorder; AN = anorexia nervosa; PTSD = posttraumatic stress disorder; MDD = major depression disorder; ANX = anxiety; SE = standard error.

a model in which psychiatric conditions and neuroticism are related to the pain conditions through the latent pain factors. Pain conditions are most highly correlated with internalizing conditions (rs = −.08. to .74), followed by externalizing conditions (rs = −.08. to .64), psychotic conditions (rs = −.20 to .28), and last, compulsive conditions (rs = −.28 to .35; Fig. 2a).

Pain-psychiatric genomic SEM

The initial pain-psychopathology model had a suboptimal fit, CFI = .883, SRMR = .078. We examined the model residuals (i.e., the difference between the observed vs. model-implied covariances) to identify pairwise covariances between conditions that were not adequately explained by the model. Among the particularly large residual covariances, several had previously reported comorbidities: cigarettes with chest pain (Wang et al., 2018), stomach pain with psychotic disorders (Dimaggio & Lysaker, 2015; E. A. Williams, 2010) and anorexia (Hildebrandt & Peyser, 2023), and IBS with both Compulsive factor conditions, OCD (Turna et al., 2019) and anorexia (Mascolo et al., 2017). Therefore, we further specified in the model and estimated residual covariances for the following condition and condition-factor pairs: CIGS with chest pain (both at baseline and during physical activity), stomach pain with schizophrenia and anorexia, and IBS with the Compulsive factor. These modifications improved the fit to CFI = .907 and SRMR = .075, which was not substantial compared with the original model. Therefore, for simpler interpretation, we did not retain these modifications and proceeded with the original model for subsequent analyses. We additionally tested two- and three-factor psychopathology models in the main model (Fig. S3 in the Supplemental Material), which further dropped the fit to CFI = .80, SRMR = .097 and CFI = .867, SRMR = .085, respectively.

The correlations among the factors in this revised model are shown in Figure 2b. General Pain showed the strongest genetic correlation with Externalizing and Internalizing (both rs = .70). General Pain correlated (r = .17) with the Psychotic disorders factor and did not significantly correlate with the Compulsive disorders factor (r = −.08, SE = .04). Furthermore, there was no overlap in 95% confidence intervals for General Pain correlations with the Externalizing and Internalizing factors compared with Thought disorder factors, underscoring the much stronger genetic relationship between the pain factor and the first two psychiatric factors. For the Musculoskeletal Pain factor, there was a weak correlation with the Externalizing factor (r = .18) and negative correlations with the remaining psychiatric factors (rs = −.23, −.32, and −.12). Despite the high correlation between the Internalizing and Externalizing factors (r = .67; Fig. S1B in the Supplemental Material), Cholesky decompositions showed that the correlation for each psychiatric factor independent of the other with General Pain was significant (.32 for both Externalizing independent of Internalizing and for Internalizing independent of Externalizing; Figs. 3a and 3b). The Cholesky decompositions also showed that after controlling for Externalizing and Internalizing factors, no positive correlations between either pain factor or the thought disorder factors remained (Fig. 3).

We additionally ran a Q-trait heterogeneity test (Grotzinger et al., 2019; Huedo-Medina et al., 2006) to determine whether a model with direct relationships between the General Pain factor and individual psychiatric conditions had a better fit than the main model (for details, see the Supplemental Material). The chi-square difference test was significant ($p = 2.1 \times 10^{-222}$, Bonferroni-corrected for 11 traits), although the CFI difference was less than .01 (.912 – .907 = .005). Given that this omnibus chi-square heterogeneity test yielded a significant difference but the change in CFI was minimal, we examined the discrepancies in the correlations predicted by the two models. Specifically, we examined the differences between standardized regression coefficients for each psychiatric condition on the General Pain factor versus the product of standardized psychiatric condition loadings onto their respective factor and that psychiatric factor's correlation with General Pain (Fig. S2 in the Supplemental Material). The average difference magnitude was .036; the highest are .077 for PTSD and .051 for anxiety (Table S2 in the Supplemental Material). Overall, these Q-trait heterogeneity analyses suggest that although the main model is not perfectly capturing all of the correlations between the pain factor and psychiatric conditions, it is capturing those correlations reasonably well (i.e., the main model's implied correlations are within .05 of the observed correlations). Thus, the data are largely consistent with a model in which the genetic correlations between pain and psychiatric conditions arise because of genetic correlations of underlying pain and psychiatric factors.

Neuroticism with pain and psychiatric conditions and factors

The full Neuroticism scale and the Depressive Affect subscale showed strong genetic correlations for a large number of musculoskeletal (r_gs = .20–.39) and visceral-pain (r_gs = .38–.50) conditions and headache (r_g = .33; Fig. 2a). The worry subscale showed low to moderate correlations (r_gs > .20 and < .40) with neck/shoulder, stomach pain, gastritis, oesophagitis, chest pain, headache, and IBS; IBS was the strongest (r_g = .53; Fig. 2a). Among psychiatric conditions, all three neuroticism scales were strongly correlated with anxiety

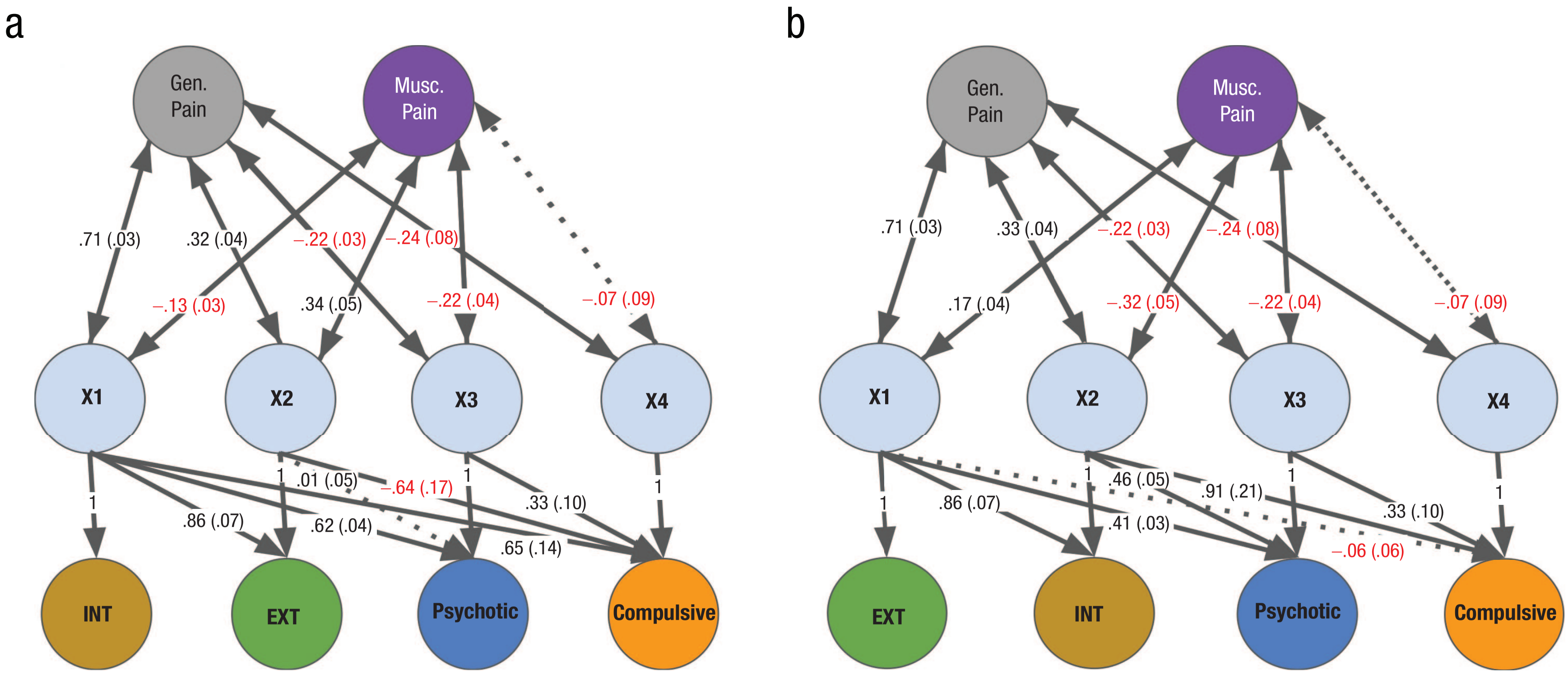

Fig. 3. Cholesky decomposition. Externalizing (EXT), Internalizing (INT), Psychotic thought (PSY), Compulsive thought (COM), General (Gen.) Pain, and Musculoskeletal (Musc.) Pain are lower-order latent factors, defined as shown in Figure S1 in the Supplemental Material available online. X1 through X4 are arbitrarily named higher-order latent constructs that capture shared and independent variance in psychiatric factors. (a) Double arrows between the pain factors and X1 represent genetic correlations between General and Musculoskeletal Pain, respectively, and the standardized genetic variance of the Internalizing factor. The double-headed arrows between the pain factors and X2 represent the genetic correlation between General Pain and Musculoskeletal Pain, respectively, and the Externalizing factor independent of the Internalizing factor. The double-headed arrows between the pain factors and X3 represent the genetic correlation between General Pain and Musculoskeletal Pain, respectively, and the Psychotic disorder factor independent of the Internalizing and Externalizing factors. (b) The decomposition is shown as in Figure 3a except that X2 partials out the genetic variance of the Externalizing factor. In both subfigures, the double arrows between the pain factors and X2 and X3 provide answers to the questions that instigated this analysis: There is shared variance between both General Pain and Musculoskeletal Pain and the Externalizing factor independent of the Internalizing factor (.33 and .34, respectively, standardized), and there is shared variance between both General Pain and Musculoskeletal Pain and the Internalizing factor independent of the Externalizing factor (.30 and −.31, respectively, standardized). In addition, there is no positive genetic correlation between either pain factor and the thought disorder factors after partialing out Externalizing and Internalizing factor genetic variance. Factor variances for the pain factors and X1 through X4 are standardized (1), and psychiatric factor variances are constrained to 0. For both models, comparative fit index = .907, standardized root mean square residual = .075, and χ^2 = 20,440(523).

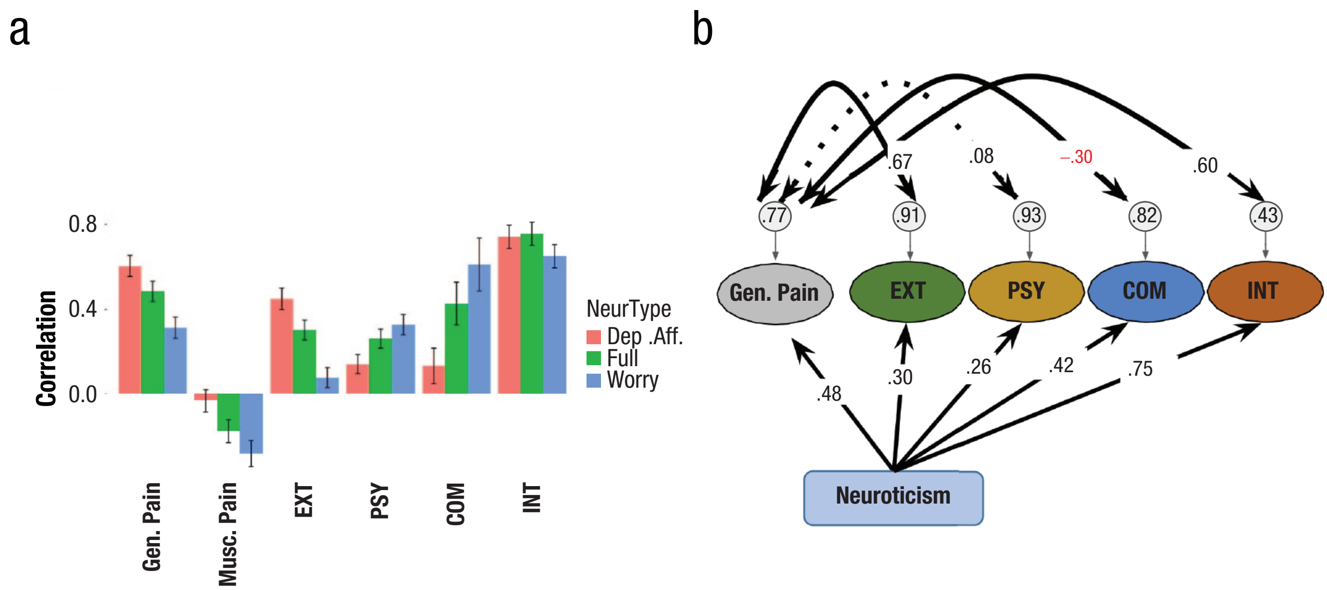

Fig. 4. (a) Correlations between neuroticism (full Neuroticism, Depressive Affect [Dep. Aff.], and Worry scales) and all factors. (b) Genomic structural equation models of neuroticism as an explanation of the pain-psychiatric factor genetic relationships. The correlations for neuroticism (full scale) with the General (Gen.) Pain factor and each of the psychiatric factors (EXT = Externalizing; INT = Internalizing; PSY = Psychotic disorder and Compulsive disorder). For Gen. Pain and INT, neuroticism explains 53% of the covariance; for Gen. Pain and EXT, neuroticism explains 19% of their covariance; for Gen. Pain and the Psychotic factor, neuroticism explains 91% of the covariance; Gen. Pain and the Compulsive factor were not significantly correlated before including neuroticism. The Musculoskeletal Pain factor was included in the model but is not shown in the figure.

(r_gs = .62–.71), depression (r_gs = .60–.69), and PTSD (r_gs = .43–.69); there was an additional robust correlation for the worry subscale and OCD (r_gs = .52; Fig. S3 in the Supplemental Material).

At the factor level, the three neuroticism scales were weakly to moderately positively correlated with the General Pain factor (Depressive Affect: r_g = .58; full scale: r_g = .48; and Worry: r_g = .26) and weakly negatively correlated with the Musculoskeletal Pain factor (Depressive Affect: r_g = −.18; Worry: r_g = −.26; full scale: *ns*; Fig. 4a). Among correlations with psychiatric factors, the neuroticism scales showed considerable heterogeneity. The highest correlations are between Internalizing and the full Neuroticism and Depressive Affect scales (r_g = .75 and r_g = .73, respectively), followed by Worry scale correlations with Psychotic and Internalizing factors (r_g = .70 and .69, respectively). The lowest correlation is between the Worry scale and the Externalizing factor.

We estimated the contribution of neuroticism to the genetic correlations between pain and psychiatric factors (Fig. 4b). Neuroticism explained the following proportions of correlation for General Pain: 53% (.53, *SE* = .03) with Internalizing disorders, 19% (.19, *SE* = .01 with Externalizing disorders, 83% (.83, *SE* = .09) with Psychotic disorders, and all of the correlation with Compulsive disorders, although this correlation was not significant to start with (Fig. 4b).

Discussion

We assessed genetic relationships between 24 chronic-pain conditions, 11 psychiatric conditions, and neuroticism at the level of latent factors, going beyond pairwise correlations between individual disorders. In addition to an important large-scale replication of prior work, we analyzed relationships between many chronic-pain conditions of different etiologies and psychiatric conditions from Internalizing, Externalizing, and Thought disorder dimensions, and we used novel genetic factors that had not been possible before a sample with the genotypic and phenotypic breadth of the UKBB was available. These factors and their correlations revealed shared underlying biological mechanisms among conditions from different domains that are not currently diagnosed or treated together or by the same specialists. With the exception of somatoform disorders, which are rare (prevalence of 2% to 3%; Konrad & Kostev, 2020; Vesterling et al., 2023) and which have been added as a spectrum of psychopathology recently and provisionally (Forbes et al., 2024; Kotov et al., 2011, 2017, 2021), within-domains focus has obfuscated recognition of comorbid, mutually exacerbating pain-psychiatric conditions, such as depression in chronic-pain patients (Jaracz et al., 2016; Katon, 1984). Currently, chronic pain is treated by primary-care practitioners or neurologists, who have limited training in mental health, and psychiatric disorders are treated by mental-health specialists, who have limited training in pain management (Darnall et al., 2016; Goesling et al., 2018). Nevertheless, there is a growing recognition of benefits to an integrated biopsychosocial pain-treatment paradigm, as evidenced by two recent articles calling for more practitioners skilled in psychology-informed pain management (Barrett, 2024; Wandner et al., 2019). Our findings offer critical evidence in support of integrated pain-psychology training for clinicians who aim to work with conditions in either domain. Specifically, the extent and pattern of genetic overlap between the examined conditions underscores the need to treat chronic pain and conditions in Externalizing and Internalizing dimensions of psychopathology as, to a large degree, linked facets of the same underlying pathophysiology.

The Internalizing and Externalizing psychopathology factors share substantial genetic risk with the General Pain factor. Furthermore, the genetic overlap between cross-condition pain and both externalizing and internalizing disorders is partly due to the shared genetic risk among all of these conditions and neuroticism. Our findings thus provide evidence for shared biological risks for chronic pain beyond direct stimulation of primary nociceptors. These risks are consistent with the biopsychosocial model of pain, a dynamic interaction between psychological, social, and pathophysiological factors (Bevers et al., 2016; Boersma et al., 2014; Gatchel, 2004; Nicholas, 2022), to which our study adds a common genetic component.

The genetic associations between the pain factors, psychiatric factors, and neuroticism in our models are consistent with previous findings for individual condition correlations. Internalizing disorders show the highest genetic correlation with cross-condition chronic pain, as previously reported (Chen et al., 2023; Johnston et al., 2019; Meng et al., 2020). It has likewise been shown that the internalizing disorders considered here have high comorbidity with chronic pain. Up to 33% of people with depression report chronic pain, and up to 75% of people with chronic pain report depression (L. S. Williams et al., 2003). Anxiety affects up to 60% of chronic-pain patients (Hooten, 2016). PTSD is also highly comorbid with chronic pain (Asmundson et al., 2002; Beckham et al., 1997; Shipherd et al., 2007). Our work advances our understanding of these comorbidities by showing substantial overlap in their genetic risks.

Externalizing conditions have substantial genetic correlations with many pain conditions. Substance use disorders in chronic-pain patients—including cannabis, alcohol, and tobacco use disorders—were recently reviewed by Martel et al. (2018). All three disorders, as defined by the fifth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (American Psychiatric Association, 2013), have a higher prevalence in chronic-pain patients. Cannabis use disorder is more common among people with chronic pain and comorbid depression (Edlund et al., 2013), anxiety (Degenhardt et al., 2015), or other mental-health problems (Ste-Marie et al., 2012). Chronic pain has a bidirectional relationship with both cigarette smoking (Ditre et al., 2011; Sanders et al., 2012; Schürks & Diener, 2008; Shiri et al., 2010) and alcohol use disorder (Cheng et al., 2000; Chopra & Tiwari, 2012; Demyttenaere et al., 2007; Ekholm et al., 2009; Lerch et al., 2003; Macfarlane & Beasley, 2015). Anxiety (Edlund et al., 2007; Kurtze et al., 1999) and depression (Edlund et al., 2007; Holmes et al., 2010) have shown higher associations with heavy alcohol use given chronic-pain comorbidities. ADHD (possibly underdiagnosed in pain patients; Kasahara et al., 2020) is associated with chronic pain in women (Asztély et al., 2019) and with increased pain sensitivity (Treister et al., 2015); neuroinflammation may be a linking mechanism (Kerekes et al., 2021).

The Psychotic (schizophrenia and bipolar disorder) and Compulsive (OCD and anorexia) thought disorder factors have almost no positive genetic correlations with chronic-pain conditions, and the small positive correlation with the Psychotic thought disorder factor is attributable to that factor's shared variance with the Internalizing and Externalizing factors (Fig. 3). This lack of independent genetic correlation suggests a lack of shared biological susceptibility between thought disorders and pain conditions. These findings are largely consistent with prior studies, which have reported substantial comorbidities between thought disorders and certain chronic-pain conditions (Birgenheir et al., 2013; Nicholl et al., 2014; Stubbs et al., 2014) but weak genetic correlations, if any. BP and SCZ have both been found to have weak genetic correlations with abdominal and multisite chronic pain (Chen et al., 2023). Although precise estimates are not provided, they appear comparable with our genetic correlation of .13 between chronic widespread pain and BP. Literature on SCZ and chronic pain includes conflicting reports. Two groups have reported that SCZ was negatively correlated with chronic pain (Owen-Smith et al., 2020) and pain intensity (Engels et al., 2014), and another study reported it to be a small risk factor for chronic pain (odds ratio = 1.02; Chen et al., 2023). Overall, earlier reports and our findings suggest that comorbidities between chronic pain and thought disorders may be environmentally mediated.

Regarding compulsive disorders, prior studies have reported these disorders to have epidemiologic but, consistent with our findings, no genetic overlap with chronic pain. Thus, anorexia is comorbid with abdominal pain (Pianucci et al., 2021) and migraine (Mustelin et al., 2014) but not genetically correlated with chronic pain (Chen et al., 2023). OCD is comorbid with chronic pain from population-based but not clinic-based studies (Fernández de la Cruz et al., 2022). To our knowledge, genetic correlations between OCD and chronic pain have not been assessed previously.

It may be of interest whether our findings would extend to other models of psychopathology. A viable alternative would be the p factor model with three correlated factors: Externalizing, Internalizing, and Thought disorder factors (Caspi et al., 2014). However, when taken into the genetic space, the constituent conditions of the Thought factor showed differential associations with other related traits in a manner consistent with two factors. Specifically, correlations for single-nucleotide polymorphisms (SNPs) (Grotzinger et al., 2022), sleep traits (Morrison et al., 2023), and pain conditions (present study) with thought disorders clearly distinguish between Psychotic and Compulsive thought disorder factors. Given this genetic heterogeneity of the original Thought disorder factor (Caspi et al., 2014), we did not test the three-factor model of psychopathology in our overall pain-psychopathology model.

Neuroticism, particularly its Depressive Affect subscale (loneliness, miserableness, and moodiness), has a strong correlation with chronic pain. The genetic overlap between cross-condition chronic pain and both the Internalizing and Externalizing factors is partly attributable to all three categories of disorders sharing genetic risk with neuroticism. This finding suggests that biological propensity for negative emotionality may explain some of the substantial co-occurrence between chronic pain and externalizing/internalizing psychopathologies. Further investigation focused on the three-way overlap between high neuroticism, chronic pain, and these psychopathologies may elucidate the mechanisms underlying pain-psychopathology comorbidity. In addition, neuroticism appears to explain much of the relationship between chronic pain and the Psychotic thought disorder factor, which together with our finding of no positive genetic correlations for chronic pain with the Psychotic thought disorder factor above and beyond Internalizing and Externalizing factors suggests minimal to no independent shared genetic risk between chronic pain and thought disorders.

At the phenotypic level, several studies have shown correlations between neuroticism and different chronic-pain conditions, such as migraine (Ashina et al., 2017), especially with depression and anxiety (Galvez-Sánchez & Montoro Aguilar, 2022); neck, shoulder, and back pain (Bru et al., 1993); and fibromyalgia (Malin & Littlejohn, 2012). One study showed a negative correlation between neuroticism and brain activity in emotional and cognitive pain-processing areas in anticipation of pain (Coen et al., 2011). Neuroticism is likewise associated with greater pain catastrophizing (Sullivan et al., 1995) and pain-related anxiety (Kadimpati et al., 2015). At the genetic level, two studies reported correlations for neuroticism with chronic pain in several of the UKBB body sites (face, head, neck/shoulder, back, stomach, pain all over the body; Chen et al., 2023; Meng et al., 2020). The latter (Chen et al., 2023) examined the shared genetic overlap in chronic and short-term pain and neuropsychiatric disorders, which included BP, SCZ, anorexia nervosa, anxiety, depression, PTSD, and ADHD. Together with both of these studies, our findings show that the association between chronic-pain-condition incidence and high neuroticism may be partly due to their shared genetic risks, which may underlie an overall susceptibility to negative sensory and emotional experiences and may or may not signify a causal relationship between neuroticism and chronic pain.

Regarding the relationship between neuroticism and health problems, there is an ongoing debate about whether neuroticism leads to higher incidence (objective experience) or higher reporting (subjective experience) of symptoms (Costa & McCrae, 1985; M. Johnson, 2003; D. Watson & Pennebaker, 1989). Our findings, albeit correlational, suggest an alternative perspective: The neuroticism-pain correlation is noteworthy in its own right. The subjective-objective dialectic around disease burden assigns a higher value to the "objective" health problem, which is deemed more worthy of medical attention. In fact, what is termed as "objective" often relies on internal symptoms experienced exclusively by the patient but measured and judged externally by the doctor. Although many conditions are accurately identified and effectively treated based on diagnostic tests, chronic pain and psychopathology are both domains of largely unseen affliction. Not unlike mental-health conditions, which are diagnosed primarily based on self-reported symptoms, chronic pain is often engendered by tissue damage, but its reported intensity shows a low correlation with tissue damage detectable in diagnostic imaging (Lee & Neumeister, 2020; Rahyussalim et al., 2020). Therefore, the extent to which they cause suffering to the patient is arguably the most important dimension of these conditions and the one that should receive attention in the clinic for mental-health and chronic-pain conditions. The substantial genetic correlation between neuroticism and chronic pain and its explanation of the chronic-pain-psychopathology relationships suggest that propensity for a negative outlook is an important factor in—and potentially treatment target for—chronic-pain pathology and comorbid psychiatric conditions.

There are several limitations in our study and attractive future directions. First, we provide a structurally informed model of genetic correlations among chronic pain, psychiatric conditions, and neuroticism. Although it is beyond the scope of this study, a functional annotation with relevant biological pathways will be an important next step to determine the specific mechanisms that underlie the genetic overlap between pain and psychiatric factors and neuroticism. Second, the source GWASs suffer from phenotype heterogeneity. Studies of conditions such as BP and SCZ included different subtypes of the umbrella psychopathology, which may have differential genetic correlations with the other conditions included in the present study. Third, the main populations used in our analyses are of White European descent, residing in Western Europe or the United States. It is our hope that as samples more representative of the world's population continue to grow in size, we will be able to check how our model performs in people with other ancestral backgrounds and from other cultural settings. Fourth, our assessment of the Externalizing dimension of psychopathology relies on substance use disorders and one neurodevelopmental disorder without a more direct externalizing-behavior measure, such as conduct problems or antisocial personality disorder, because there is not a sufficiently sized GWAS of these measures available. Fifth, the post hoc addition of four correlations of residuals, although theoretically sound and supported in the literature, as mentioned above, will need to be validated in an independent data set to check for false-positive relationships and overfitting.

In conclusion, our results reconceptualize chronic pain in light of its genetic overlap with neuroticism and psychiatric conditions, especially those of Internalizing and Externalizing dimensions of psychopathology, and underscore the importance of holistic clinical assessments. Our findings further suggest that treatments may benefit from considering frequently comorbid conditions and their shared genetic risk factors.

Transparency

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Declaration of Conflicting Interests

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